

Notes: Li, Shen & Zhang (RFS, 2024)
Local Effects of Global Capital Flows: A China Shock in the U.S. Housing Market

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1 Big picture

- **Question.** What are the local real effects of foreign residential real estate capital inflows? In particular, what happened when Chinese buyers sharply increased home purchases in California after 2008?
- **Setting.** The paper studies **single-family housing transactions** in the **three largest California CBSAs**, covering **17 counties** and **773 ZIP codes** from **2001 to 2013**. Outcomes are measured at the **ZIP code-year** level.
- **Core measurement challenge.** U.S. deed records do not report the buyer's country of origin, so the paper must first construct a measure of foreign real estate capital inflows from transaction data.
- **Main measurement contribution.** The authors build a **three-step imputation algorithm** that combines ethnic-name matching, an all-cash filter, and an adjustment for resident non-Anglo buyers to isolate **nonresident foreign purchases**. This gives two continuous treatment measures: **Chinese housing transaction value (CHTV)** and **Chinese housing transaction count (CHTC)**.
- **Two key facts.** First, there is a **China shock** in U.S. housing: foreign Chinese purchases surge after 2008 and reach about **3.5%** of total California housing transaction value by 2013. Second, there is strong **home bias**: these purchases concentrate in ZIP codes with historically larger ethnic Chinese populations.
- **Main theoretical mechanism.** A positive foreign housing demand shock raises local house prices. That boosts homeowners' housing net worth and increases demand for local *nontradable* goods and employment. But the same price increase can also displace local residents. The paper's model emphasizes the tension between a **housing net worth channel** and a **displacement channel**.
- **Main empirical design.** The paper combines a **continuous-treatment difference-in-differences** design with an **IV strategy**. The instrument is **historical ethnic Chinese population share in a ZIP code** interacted with **aggregate Chinese purchases in California**, exploiting the documented home-bias pattern.
- **Main employment result.** A **1%** increase in foreign Chinese housing transaction value raises local employment by **0.124%**; a **1%** increase in transaction count raises it by **0.247%**. The number of establishments also rises by **0.128%** and **0.236%**, respectively.
- **Main house-price result.** A **1%** increase in Chinese transaction value raises the local Zillow home-value index by **0.108%**; the count-based elasticity is **0.195%**. Using transaction prices instead of Zillow, the elasticities are **0.145%** and **0.251%**.
- **Main displacement result.** A **1%** increase in Chinese transaction value lowers the number of local tax filers by **0.035%**; the count-based estimate is **0.061%**. So the inflows stimulate employment but also push residents out.
- **Key sectoral result.** The employment gains are concentrated in **nontradable sectors**: the elasticities are **0.160%** for CHTV and **0.282%** for CHTC. Tradable-sector employment does

not respond robustly.

- **Key distributional result.** The displacement effect is concentrated among **low-income residents**. The number of low-income tax returns falls by **0.063%** to **0.112%** per 1% increase in Chinese purchases, while high-income households are not significantly displaced.
- **Main credibility result.** The event-study coefficients are flat before 2008, the reverse-causality tests are null, the first stage is strong, and the results survive post-crisis subsamples and controls for foreclosures, all-cash transactions, and local financial-sector exposure.
- **Economic takeaway.** Foreign housing capital inflows can **raise local employment and house prices at the same time**. The paper's central message is that real estate inflows can simultaneously act like a local demand stimulus and a gentrification force.

2 Data and measurement (institutional context and key objects)

2.1 Why this setting is hard to study and why California is useful

- Official U.S. data on foreign housing purchases are weak. County deed offices do *not* record buyers' countries of origin, and the National Association of Realtors relies on low-response surveys.
- The paper therefore turns to **transaction-level housing records** and builds its own measure of foreign purchases.
- California is a natural laboratory because the paper uncovers a very large post-2008 surge in Chinese purchases there, and because the state contains neighborhoods with very different historical ethnic Chinese settlement patterns.

Interpretation. The paper's empirical contribution starts before the regressions begin: it first solves a measurement problem that normally prevents researchers from observing residential capital inflows directly.

2.2 Real estate transaction data and sample construction

- The core housing data come from **DataQuick**, which compiles records from county deed and assessor offices.
- For each transaction, the data include the sale price, closing date, address, home characteristics, financing information, and buyer and seller names.
- The analysis focuses on **single-family residential homes**.
- The sample spans **2001–2013** in the **three largest California core-based statistical areas**, covering **17 counties** and **773 ZIP codes**.

- Both resident- and nonresident-initiated transactions appear in the raw data.

Interpretation. The transaction records are rich enough to identify who bought, how they financed the purchase, and where the property is located. That is exactly what the paper needs to build the foreign-purchase measures.

2.3 Step 1 of the imputation algorithm: ethnic name matching

- The paper applies the **Kerr** ethnic name-matching algorithm to buyers' first and last names.
- The algorithm classifies buyers into **eight ethnic groups**: Anglo-Saxon/English, Chinese, European, Hispanic, Indian, Japanese, Korean, and Russian.
- The match rate is **97%**.
- In the first step, the paper keeps only transactions where the buyer is assigned to one of the seven non-Anglo groups with **probability one**.

Interpretation. This step is deliberately conservative. The authors prefer to lose some observations rather than classify ambiguous names incorrectly.

2.4 Step 2: using the all-cash filter to identify likely foreign buyers

- Foreign nonresidents have limited access to the U.S. mortgage market.
- The paper therefore keeps only **all-cash transactions** made by non-Anglo buyers.
- The intuition is backed by outside evidence: the NAR reports that **76%** of nonresident foreign buyers paid all cash, versus **33%** of resident foreign buyers.

Interpretation. The all-cash filter is a proxy for *nonresident* status. It likely understates the total size of foreign demand, but it increases measurement accuracy for the part the paper wants to isolate most cleanly.

2.5 Step 3: removing resident non-Anglo purchases

- Even among all-cash non-Anglo purchases, some buyers may be **U.S. residents** rather than nonresident foreigners.
- The paper adjusts for this by subtracting the component attributable to resident non-Anglo buyers, using Anglo-Americans' all-cash propensity as the benchmark for lawful residents with standard mortgage access.
- The adjusted foreign transaction value for ethnic group f in ZIP code z and year t is

$$fHTV_{zt} = \left[HTV_{fzt}^{cash} - \frac{HTV_{Azt}^{cash}}{HTV_{Azt}} \times HTV_{fzt} \right] \frac{HTV_{Azt}}{HTV_{Azt} - HTV_{Azt}^{cash}}, \quad (1)$$

where A denotes Anglo-Americans. The same construction gives $fHTC$, the adjusted foreign transaction count.

Interpretation. The final measure is trying to answer a very specific question: *how much of the housing demand in a ZIP-year comes from nonresident foreigners of ethnicity f ?*

2.6 The main treatment variables

The paper then specializes the foreign-purchase measures to **Chinese** buyers:

- **CHTV**: Chinese housing transaction value.
- **CHTC**: Chinese housing transaction count.
- Regressions typically use their logs, $\ln(CHTV)$ and $\ln(CHTC)$.

Interpretation. Value captures the intensive margin; count captures the extensive margin. When count effects are at least as large as value effects, it is harder to argue that the results are driven only by a few expensive luxury purchases.

2.7 Two stylized facts: China shock and home bias

- **Fact 1: China shock.** Foreign Chinese purchases rise sharply after **2008**. By 2013, they account for about **3.5%** of total California housing transaction value, more than a **30-fold** increase relative to the pre-2008 level.
- The count-based series looks very similar to the value-based series, suggesting Chinese buyers were purchasing across the housing-quality distribution rather than only top-end properties.
- **Fact 2: home bias.** Chinese purchases are strongly concentrated in ZIP codes with higher **historical ethnic Chinese population share**.
- By 2013, the foreign Chinese transaction share in ZIP codes in the **top two deciles** of historical Chinese population share exceeds that in the **bottom eight deciles** by roughly **6 percentage points**.
- In absolute terms, foreign Chinese purchases reach about **7%** of total transaction value and count in the top-two-decile ZIP codes.

Interpretation. These two facts generate the paper's identifying variation. The time series gives the shock, and the cross section gives differential exposure to the shock.

2.8 Outcome variables and controls

The main outcomes are:

- **Total employment and number of establishments**, from **Census Bureau ZIP Codes Business Patterns**.

- **Nontradable-sector employment** and **tradable-sector employment**, based on a NAICS-sector classification from Mian, Rao, and Sufi (2013).
- **House prices**, measured both by the **Zillow Single-Family Home Value Index** and by transaction prices from DataQuick.
- **Number of tax filings**, from the **IRS**, used as a proxy for the local resident count.
- **Average household income** and, later, tax returns split by income group.

The main controls include:

- pre-sample **population**,
- **population density**,
- **education** (share with a bachelor’s degree),
- an indicator for whether there is a **college within five miles**,
- pretrends in **income**,
- pretrends in the relevant **outcome variable**,
- and **county-year fixed effects**.

Interpretation. County-year fixed effects are extremely important here: they force identification to come only from **within-county cross-ZIP** variation, which absorbs county-level shocks such as broad local economic booms or county-wide housing policies.

2.9 Summary statistics and scale of the shock

- Across all ZIP code-years, average Chinese transaction value is about **\$2.43 million** and average Chinese transaction count is **5.53**.
- In the **pre-shock period** (2001–2007), the average ZIP code sees about **\$1.04 million** of Chinese purchases and **1.69** Chinese transactions per year.
- In the **post-shock period** (2008–2013), these averages rise to about **\$3.11 million** and **7.43** transactions.
- The paper emphasizes a useful contrast: foreign Chinese purchases surge even though **overall housing transactions decline** relative to the pre-shock period.
- The difference-in-difference change in the log treatment variable between top-two-decile and bottom-eight-decile ZIP codes is about **0.36** for $\ln(CHTV)$ and **0.13** for $\ln(CHTC)$.

Interpretation. The shock is economically large, spatially uneven, and not just a mirror image of a general housing boom.

3 Models and empirical specifications (all equations + interpretation)

3.1 Conceptual framework: two regions, two sectors, fixed housing

- The model has **two regions**, indexed by $z \in \{0, 1\}$.
- Each region produces a **nontradable good** and a **tradable good**.
- Housing supply is fixed at H in each region.
- Workers can switch sectors within a region and can also **commute across regions**.
- Households have Cobb–Douglas preferences over nontradables, tradables, and housing.

Interpretation. The model is designed to isolate the local-demand consequences of a housing demand shock. Fixed housing supply makes price effects immediate and transparent.

3.2 Initial equilibrium

In the symmetric initial equilibrium, there is no commuting, wages are equalized across sectors, and the economy is locally self-contained:

$$P_z^{N*} = \frac{b}{a}, \quad P_z^{T*} = 1, \quad P_z^{H*} = \frac{1 - \alpha - \beta}{\alpha + \beta} \frac{b}{H},$$

$$w_z^{N*} = w_z^{T*} = b, \quad D_z^* = \frac{b}{\alpha + \beta}, \quad e_z^{N*} = \frac{\alpha}{\alpha + \beta}, \quad e_z^{T*} = \frac{\beta}{\alpha + \beta}.$$

Interpretation. Before the foreign-demand shock, house prices, employment shares, and incomes are the same across regions.

3.3 The foreign housing demand shock

Region 1 then receives an exogenous foreign demand shock:

$$H = C_1^H + C_f^H,$$

where C_f^H is housing demand from foreigners.

The resulting housing-market equilibrium implies

$$P_1^H = \frac{(1 - \alpha - \beta)b(1 - \delta)}{(\alpha + \beta)H - C_f^H}, \tag{2}$$

$$P_0^H = \frac{(1 - \alpha - \beta)b(1 + \delta)}{(\alpha + \beta)H}. \tag{3}$$

Interpretation. Foreign demand raises house prices most directly in Region 1, the neighborhood that receives the capital inflow. But the model also allows spillovers to Region 0 through commuting and relocation.

3.4 Commuting as the displacement margin

- Workers in Region 1 can choose to live in Region 0 and commute if housing becomes too expensive in Region 1.
- Commuting cost is heterogeneous across workers.
- A larger foreign housing demand shock increases the fraction of workers who commute away from the directly affected region.

Interpretation. In the model, **displacement** is not just a loose idea. It is literally the margin by which workers stop residing in the expensive region even if they still work there.

3.5 The key decomposition: housing net worth versus displacement

The equilibrium level of nontradable employment in Region 1 can be written as

$$e_1^N = \frac{\alpha H}{b} P_1^H + \alpha(1 - \lambda\delta), \quad (4)$$

and in Region 0 as

$$e_0^N = \frac{\alpha H}{b} P_0^H + \alpha(1 + \lambda\delta). \quad (5)$$

The first term captures the **housing net worth channel**; the second captures the **displacement/commuting channel**.

Interpretation. A foreign housing demand shock can raise nontradable employment because higher house prices expand homeowners' balance sheets. But it can also reduce local demand if residents get pushed out. The empirical question is which force dominates.

3.6 The model's qualitative predictions

The theory delivers four clear predictions:

1. foreign housing demand should **raise total employment** in the directly affected region;
2. it should **raise house prices**;
3. it should also **increase displacement** of local residents;
4. the employment effect should be strongest in the **nontradable sector**, where local demand matters most.

Interpretation. The model is not meant to fit every detail of the data. It is a guide for how to read the empirical patterns sector by sector.

3.7 Baseline DID-IV specification

The paper's core empirical specification is

$$\ln(Y_{zt}) = \alpha + \theta \ln(CHTV_{zt}) + \beta \ln(CHTV_{zt}) \times \mathbf{1}\{t \geq 2008\} + \gamma X_{z,0} + \eta_{ct} + \varepsilon_{zt}, \quad (6)$$

$$\ln(Y_{zt}) = \tilde{\alpha} + \tilde{\theta} \ln(CHTC_{zt}) + \tilde{\beta} \ln(CHTC_{zt}) \times \mathbf{1}\{t \geq 2008\} + \tilde{\gamma} X_{z,0} + \tilde{\eta}_{ct} + \tilde{\varepsilon}_{zt}. \quad (7)$$

- Y_{zt} is the ZIP-year outcome of interest.
- $X_{z,0}$ contains the pre-sample controls.
- η_{ct} are county-year fixed effects.
- The coefficients of interest are β and $\tilde{\beta}$.

Interpretation. The specification is a **generalized DID with continuous treatment**. It compares ZIP codes that received more Chinese purchases with ZIP codes that received fewer, before and after the 2008 China shock.

3.8 The IV strategy

The paper instruments Chinese purchases with

$$CHShare_{z,0} \times CHTTV_t \quad \text{and} \quad CHShare_{z,0} \times CHTTC_t,$$

where:

- $CHShare_{z,0}$ is the historical ethnic Chinese population share in ZIP code z from the **2000 Census**,
- $CHTTV_t$ and $CHTTC_t$ are total Chinese transaction value and count in California in year t .

Interpretation. This is essentially a **Bartik-style exposure design**. The aggregate time-series shock is the statewide rise in Chinese purchases; local exposure is the ZIP code's preexisting Chinese population share. The home-bias fact is what makes the instrument economically sensible.

3.9 Event-study specification

To test dynamic effects and the parallel-trends assumption, the paper estimates

$$\ln(Y_{zt}) = \alpha + \theta \ln(CHTV_{zt}) + \sum_{k=2001}^{2013} \beta_k \ln(CHTV_{zt}) \times \mathbf{1}\{t = k\} + \gamma X_{z,0} + \eta_{ct} + \varepsilon_{zt}, \quad (8)$$

$$\ln(Y_{zt}) = \tilde{\alpha} + \tilde{\theta} \ln(CHTC_{zt}) + \sum_{k=2001}^{2013} \tilde{\beta}_k \ln(CHTC_{zt}) \times \mathbf{1}\{t = k\} + \tilde{\gamma} X_{z,0} + \tilde{\eta}_{ct} + \tilde{\varepsilon}_{zt}, \quad (9)$$

using **2007** as the omitted year.

Interpretation. If the paper’s design is credible, the pre-2008 interaction coefficients should be close to zero, and the post-2008 coefficients should turn positive exactly when the China shock begins.

3.10 Reverse-causality test

The paper also regresses *pre-shock* employment and income changes on *post-2008* Chinese purchase intensity:

$$Y_{pre,z} = \alpha_0 + \beta_0 \ln(CHTV_{08-13,z}) + \gamma_0 X_z + \eta_c + \varepsilon_z, \quad (10)$$

$$Y_{pre,z} = \tilde{\alpha}_0 + \tilde{\beta}_0 \ln(CHTC_{08-13,z}) + \tilde{\gamma}_0 X_z + \tilde{\eta}_c + \tilde{\varepsilon}_z. \quad (11)$$

Interpretation. This is a simple but useful falsification exercise: if neighborhoods that later received Chinese inflows were already on systematically better trajectories, the coefficients here should be positive. The paper mostly finds they are not.

4 Identification and empirical design (what makes the estimates credible)

4.1 What identifies the estimates

- The design exploits **two dimensions of variation** at once:
 - a **time-series break** after 2008 in Chinese purchases,
 - and **cross-sectional exposure** based on historical Chinese settlement patterns.
- County-year fixed effects mean the paper is not comparing Silicon Valley to Los Angeles in levels. It is comparing **ZIP codes within the same county-year** that were more versus less exposed to the China shock.
- The instrument uses historical ethnic Chinese population share only as an **exposure shifter**, not as a direct measure of economic activity.

Interpretation. The design is strongest when one thinks of the 2008 China shock as a common demand shift whose local intensity was shaped by preexisting social geography.

4.2 Why the home-bias instrument is plausible

- Chinese buyers disproportionately choose neighborhoods with more historical ethnic Chinese residents.
- That home bias gives the first stage real economic content: places with higher Chinese population share are exactly where the statewide rise in Chinese purchases should land more strongly.

- The first-stage F -statistics are strong throughout the paper, typically ranging from about **22** to **65**.

Interpretation. The IV is not a mechanical shift-share device with a mysterious first stage. The paper first documents the home-bias fact directly in the raw data, then uses it for identification.

4.3 Main threats and how the paper addresses them

- **Neighborhood growth potential.** Chinese buyers might target places that were already going to boom.
- **Crowding out of domestic buyers.** Local buyers priced out of Chinese-targeted ZIP codes might spill into nearby ZIP codes, biasing the estimates.
- **Immigration confounds.** Historical Chinese enclaves might also receive more migrants from China after 2008.
- **County-level booms or land policy.** For example, a tech boom or a restrictive county land policy might drive both house prices and employment.

The paper's answers:

- it includes **pre-sample controls** and pretrends,
- it includes **county-year fixed effects**,
- it shows **no large immigration surge** from China in the relevant period,
- and it shows the employment gains are concentrated in **nontradables**, which is harder to reconcile with a pure technology-boom story.

Interpretation. None of these checks is individually decisive, but together they make it much harder to read the results as simple omitted-variable bias.

4.4 Parallel trends and the event-study evidence

- In Figure 5, the coefficients on Chinese-purchase exposure for **total employment** are close to zero before 2008 and then become positive afterward.
- The same pattern is even clearer for **nontradable employment**.
- For **tradable employment**, the dynamic coefficients do not show a stable post-shock rise.
- The paper reports a similar event-study pattern for **house prices** in the Internet Appendix.

Interpretation. This is the paper's most transparent design check. The data do not show a steady pre-2008 drift in the treated ZIP codes that would have foreshadowed the results.

4.5 Reverse causality and placebo-style evidence

- Table 5 asks whether ZIP codes that later attracted more Chinese inflows already had stronger **employment** or **income** growth in the pre-shock period.
- The answer is essentially no: the coefficients are small and statistically insignificant.
- The Internet Appendix also reports a balance-style test showing that the historical Chinese-share exposure is not strongly correlated with pre-shock outcomes.

Interpretation. The paper cannot prove foreign Chinese buyers were randomly assigned across space, but it does show they were not simply chasing neighborhoods with visibly stronger preexisting economic momentum.

4.6 Financial-crisis confounds

- Because the China shock begins around the global financial crisis, one worry is that high-Chinese-share neighborhoods were simply affected differently by the crisis.
- The paper therefore reruns the main regressions in the **post-crisis period (2012–2013)** and still finds strong effects.
- It also controls separately for:
 - local **foreclosure shares**,
 - local **all-cash transaction shares**,
 - local **financial-sector employment shares**.
- The baseline employment and house-price results remain similar.

Interpretation. These checks matter a lot because the timing of the shock overlaps with a period when housing markets were moving for many reasons other than foreign demand.

4.7 What the design can and cannot distinguish

- The estimates are **local general equilibrium** effects, not national aggregate effects.
- The paper focuses on **nonresident foreign capital inflows**; it may understate total foreign demand because it excludes some financed purchases and most corporate purchases.
- The paper does not deliver a full **welfare analysis**. It shows who gains and who gets displaced, but it does not fully weigh homeowners, renters, incumbent businesses, and newcomers against each other.
- It also cannot observe migration flows perfectly, so it proxies displacement with **tax-return counts**.

Interpretation. The paper is strongest on *local real effects and mechanisms*; it is more limited on aggregate incidence and welfare.

4.8 Relation to the literature

- Relative to the foreign-housing-demand literature, the paper goes beyond **house prices** and measures **real economic effects**.
- Relative to Mian and Sufi-style work, it studies a **positive housing demand shock** rather than a housing crash, but still emphasizes household balance sheets and nontradable demand.
- Relative to the China-shock trade literature, it studies a **finance-side China shock** rather than an import-competition shock.

Interpretation. One of the paper's biggest conceptual contributions is to reframe foreign real estate purchases as a form of international capital flow with macro-real consequences, not just as a housing-market curiosity.

5 Tables and figures

5.1 Figures 1–3: the China shock and home bias

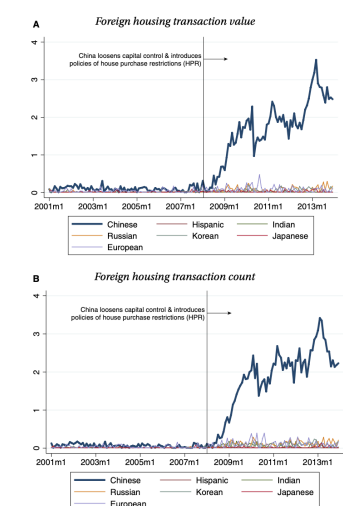


Figure 1
Foreign home purchases in the Californian real estate market
 Panel A plots the share of total housing transactions by dollar value made by seven groups of foreigners in the three largest core-based statistical areas (CBSAs) in California. Panel B plots the analog for the share of housing transactions by count. The vertical line marks the timing of China's policies of capital control, housing, and housing purchase restrictions. The sample period runs monthly from 2001 to 2013. Source: DataQuick and authors' calculations.

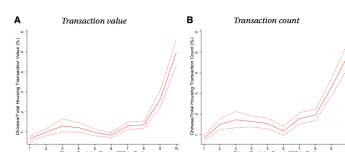


Figure 2
Foreign Chinese housing purchases versus historical ethnic Chinese population
 Panel A shows foreign Chinese housing transaction share, as measured by value, between 2008 and 2013 as a function of historical ethnic Chinese population at the ZIP code level by decade based on the 2000 Census Bureau Survey. Panel B shows the analog for foreign Chinese housing transaction share, as measured by count. The dotted lines denote the 95% confidence intervals based on 1,000 bootstrapping samples. Source: DataQuick, Census Bureau, and authors' calculations.

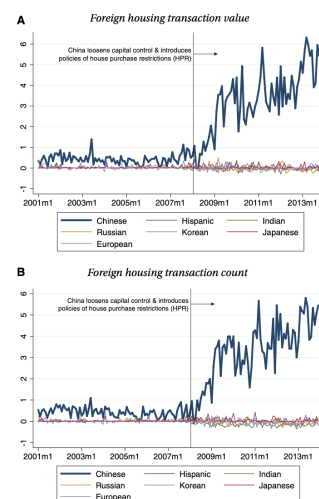


Figure 3
Foreign home purchases in the Californian real estate market: Difference between ZIP codes in top two versus bottom eight deciles of the ethnic Chinese population share
 Panel A plots the difference in foreign housing transaction share, as measured by value, between ZIP codes in the top two and bottom eight deciles of the ethnic Chinese population (based on the 2000 Census Bureau survey) in the three largest core-based statistical areas (CBSAs) in California. Panel B plots the analog for foreign housing transaction share by count. The vertical line marks the timing of China's policies of capital control, housing, and housing purchase restrictions. The sample period runs monthly from 2001 to 2013. Source: DataQuick and authors' calculations.

Read:

- Figure 1 shows that after 2008, foreign Chinese purchases jump sharply while purchases by other foreign groups remain tiny.
- The Chinese series reaches about **3.5%** of California housing transaction value by 2013.
- Figure 2 shows a sharply increasing relationship between foreign Chinese purchase intensity and **historical ethnic Chinese population share**.

- Figure 3 turns that cross section into a simple treatment-versus-control picture: top-two-decile Chinese ZIP codes pull away from the rest after 2008 by about **6 percentage points**.

Economic message. These figures do most of the paper’s heavy lifting visually. They show both that there really was a Chinese demand shock and that some ZIP codes were mechanically much more exposed to it than others.

5.2 Figure 4: the model’s comparative statics

Read:

- The stylized diagram shows that a positive foreign housing demand shock raises **house prices** and **total employment** in the directly affected region.
- **Nontradable employment** rises as long as the housing net worth channel dominates the displacement channel.
- **Tradable employment** is ambiguous because commuting can partially offset the local labor reallocation.
- The number of **commuters** rises with the shock.

Economic message. The figure is a roadmap for the empirical sections. Once one sees a positive employment response together with a positive house-price response and a negative resident-count response, the data line up exactly with the model’s channel decomposition.

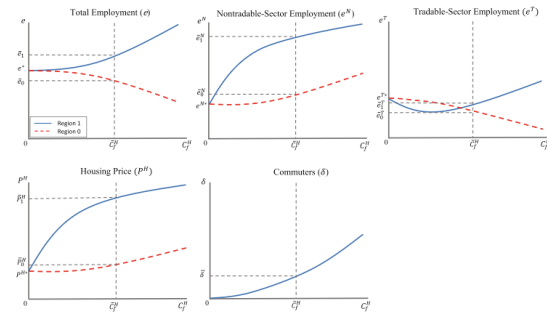


Figure 4
Local effects of a positive foreign housing demand shock, stylized diagram
This figure provides a stylized illustration of the effects of a positive foreign housing demand shock (C_F^H) on local total employment (e), nontradable-sector employment (e^N), tradable-sector employment (e^T), house prices (P^H), and number of commuters (δ) in Regions 1 (the directly impacted region) and 0. The lines show the comparative statics across different equilibria under different amounts of foreign housing demand shock, where the blue solid line denotes the set of equilibria for Region 1 and the red dotted line denotes that for Region 0. When there is no foreign housing demand shock, $C_F^H=0$, the local economy for both regions is at the initial equilibrium, with total employment at e^* , nontradable-sector employment at e^{N*} , tradable-sector employment at e^{T*} , housing price at P^{H*} , and number of commuters at zero. The gray dotted lines plot the equilibrium under a positive foreign demand shock of magnitude $\bar{C}_F^H$. The total employment in Regions 1 and 0 is $\bar{e}_1$ and $\bar{e}_0$, respectively, nontradable-sector employment is $\bar{e}_1^N$ and $\bar{e}_0^N$, tradable-sector employment is $\bar{e}_1^T$ and $\bar{e}_0^T$, housing price is $\bar{P}_1^H$ and $\bar{P}_0^H$, and the number of commuters is $\bar{\delta}$.

Table 1
Summary statistics of key variables

	All periods (2001–2013)		Preshock period (2001–2007)		Postshock period (2008–2013)	
	Mean	Std. dev.	Mean	Std. dev.	Mean	Std. dev.
	(1)	(2)	(3)	(4)	(5)	(6)
Foreign Chinese transaction						
CHTV (\$)	2.43M	4.95M	1.04M	1.54M	3.11M	5.83M
CHTC	5.53	10.69	1.69	2.22	7.43	12.55
ln(CHTV)	13.76	1.40	13.24	1.23	14.02	1.46
ln(CHTC)	0.79	1.30	0.11	0.88	1.13	1.34
Total housing transaction						
Value (\$)	138.18M	98.31M	188.65M	120.33M	113.13M	73.40M
Count	303.89	224.95	372.44	249.63	269.85	203.26
Top-two-decile Chinese ZIP codes						
ln(CHTV)	14.50	1.34	13.83	1.18	15.01	1.23
ln(CHTC)	1.15	1.30	0.47	1.06	1.67	1.23
Bottom-eight-decile Chinese ZIP codes						
ln(CHTV)	13.45	1.31	12.86	0.91	13.68	1.37
ln(CHTC)	0.64	1.27	-0.12	0.65	0.95	1.33
Outcome variables						
Log total employment	9.20	0.95	9.31	0.90	9.15	0.97
Log nontradable employment	8.11	0.90	8.21	0.84	8.06	0.92
Log tradable employment	5.95	1.93	6.17	1.94	5.84	1.92
Log establishment number	6.53	0.76	6.59	0.70	6.51	0.79
Log house price	13.21	0.59	13.19	0.55	13.22	0.61
Log number of tax filings	9.63	0.55	9.61	0.50	9.64	0.56
Log household income	11.07	0.53	11.05	0.51	11.08	0.54

This table presents the summary statistics of the key variables in the data set at the ZIP code-year level. Columns 1–2 show the annual means and standard deviations across ZIP codes for the whole sample period (2001–2013). Columns 3–4 show the summary statistics for the pre-China shock period (2001–2007), and columns 5–6 for the post-China shock period (2008–2013). Foreign Chinese housing transaction value (CHTV) and count (CHTC) are constructed using the three-step imputation algorithm described in Section 1.2. “Top-two-decile Chinese ZIP codes” refer to ZIP codes in the top two deciles of the historical ethnic Chinese population share—ZIP codes that received more foreign Chinese housing capital inflows. “Bottom-eight-decile Chinese ZIP codes” refer to those in the bottom eight deciles of the historical ethnic Chinese population share—ZIP codes that received less foreign Chinese housing capital inflows. Source: DataQuick, Zillow, Census Bureau, Internal Revenue Service, and authors’ calculations.

5.3 Table 1: summary statistics

Read:

- The average ZIP code sees Chinese transaction value rise from roughly **\$1.04 million** in 2001–2007 to **\$3.11 million** in 2008–2013.
- The average count rises from **1.69** to **7.43**.
- The treated ZIP codes—those in the top two deciles of historical Chinese population share—see much larger increases in log Chinese purchases than the comparison ZIP codes.
- Mean log employment and house prices do not jump mechanically in the raw summary statistics, suggesting the action is in the differential changes across ZIP codes.

Economic message. The table shows that the China shock is quantitatively large even before any econometrics, and that it is not simply tracking the overall housing market cycle.

5.4 Table 2: total employment and establishments

Read:

- Columns 1–2: a 1% increase in Chinese purchases raises **total employment** by **0.124%** (value) and **0.247%** (count).
- Columns 3–4: the number of **establishments** rises by **0.128%** and **0.236%**.
- The coefficients on the non-interacted Chinese-purchase variables are small and insignificant, so the treated neighborhoods do not look systematically different before the shock.
- First-stage F -statistics range from about **30** to **59**.

Economic message. Foreign housing capital inflows do not just revalue assets; they also generate enough extra local demand to increase business formation and employment.

Table 2
Foreign Chinese housing demand and total employment

	Total Employment		Number of Establishments	
	(1)	(2)	(3)	(4)
$\ln(CHTV) \times Post$	0.124** (0.060)		0.128*** (0.045)	
$\ln(CHTV)$	0.043 (0.092)		−0.014 (0.070)	
$\ln(CHTC) \times Post$		0.247*** (0.095)		0.236*** (0.074)
$\ln(CHTC)$		−0.024 (0.112)		−0.077 (0.086)
Controls	Yes	Yes	Yes	Yes
County-year FE	Yes	Yes	Yes	Yes
First-stage F -statistic	56	30	59	30
Obs.	4,166	4,166	4,166	4,166

This table reports regression results from Equations (7a) and (7b) with log total employment size (columns 1–2) and log number of establishments (columns 3–4) as dependent variables. $CHTV$ ($CHTC$) denotes foreign Chinese housing transaction value (count) instrumented by the weighted aggregate Chinese housing transaction value (count) in California, with the preexisting shares of ethnic Chinese population across ZIP codes as weights. $Post$ is an indicator variable that takes the value 1 if the year is 2008 or after, and 0 otherwise. All regressions control for pre-sample period ZIP code-level population, population density, education (share of population with a bachelor's degree), an indicator variable for whether there is a college within a five-mile distance, and pretrends of income (1998–2001) and of the outcome variable (1996–2000). Standard errors, shown in parentheses, are clustered at the ZIP code level. * $p < .1$; ** $p < .05$; *** $p < .01$.

Table 3
Foreign Chinese housing demand, local home prices, and displacement

	Home Prices (Zillow)		Home Prices (Transactions)		Number of Tax Returns	
	(1)	(2)	(3)	(4)	(5)	(6)
$\ln(CHTV) \times Post$	0.108*** (0.021)		0.145*** (0.019)		−0.035*** (0.013)	
$\ln(CHTV)$	−0.030 (0.027)		−0.058** (0.024)		0.021 (0.017)	
$\ln(CHTC) \times Post$		0.195*** (0.042)		0.251*** (0.043)		−0.061*** (0.022)
$\ln(CHTC)$		−0.084** (0.040)		−0.125*** (0.037)		0.039* (0.024)
Controls	Yes	Yes	Yes	Yes	Yes	Yes
County-year FE	Yes	Yes	Yes	Yes	Yes	Yes
First-stage F -stat.	58	25	65	27	53	22
Obs.	3,893	3,893	4,152	4,152	3,927	3,927

This table reports regression results from Equations (7a) and (7b) with log Zillow Single-Family Home Value Index (columns 1–2), log housing transaction values from DataQuick (columns 3–4), and log number of tax filers (columns 5–6) as dependent variables. $CHTV$ ($CHTC$) denotes foreign Chinese housing transaction value (count) instrumented by the weighted aggregate foreign Chinese housing transaction value (count) in California, with the preexisting shares of ethnic Chinese population across ZIP codes as weights. $Post$ is an indicator variable that takes the value 1 if the year is 2008 or after, and 0 otherwise. All regressions control for pre-sample period ZIP code-level population, population density, education (share of population with a bachelor's degree), an indicator variable for whether there is a college within a five-mile distance, and pretrends of income (1998–2001) and of the outcome variable (1996–2000). Columns 3–4 additionally control for home characteristics, including number of bathrooms, square footage, and age of the home. * $p < .1$; ** $p < .05$; *** $p < .01$.

5.5 Table 3: house prices and displacement

Read:

- Zillow home values rise by **0.108%** per 1% increase in Chinese transaction value and by **0.195%** per 1% increase in Chinese transaction count.
- Transaction-price regressions yield slightly larger elasticities: **0.145%** and **0.251%**.
- The number of tax returns falls by **0.035%** and **0.061%**, respectively.

Economic message. This is the paper's core mechanism table. The same foreign inflows that raise local housing wealth also reduce the number of local residents.

5.6 Table 4: sectoral employment and average income

Read:

- Nontradable employment rises by **0.160%** (value) and **0.282%** (count).
- Tradable employment is not estimated precisely and does not display a robust positive effect.
- Even after removing construction from nontradables, the coefficients remain positive at **0.103%** and **0.190%**.
- Average household income rises by **0.073%** and **0.114%**.

Economic message. The employment response is exactly where the model says it should be: in local-service sectors. The rise in average income also hints that the displacement margin is reshaping neighborhood composition.

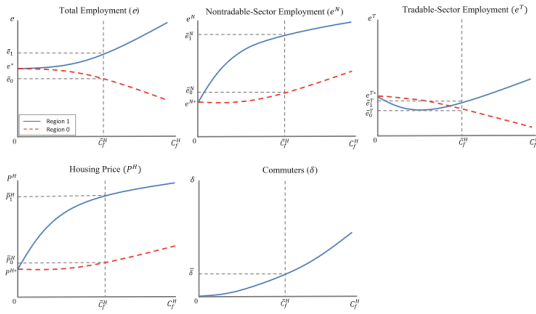


Figure 4
Local effects of a positive foreign housing demand shock, stylized diagram
This figure provides a stylized illustration of the effects of a positive foreign housing demand shock (C_f^H) on local total employment (e), nontradable-sector employment (e^N), tradable-sector employment (e^T), house prices (p^H), and number of commuters (δ) in Regions 1 (the directly impacted region) and 0. The lines show the comparative statics across different equilibria under different amounts of foreign housing demand shock, where the blue solid line denotes the set of equilibria for Region 1 and the red dotted line denotes that for Region 0. When there is no foreign housing demand shock, $C_f^H = 0$, the local economy for both regions is at the initial equilibrium, with total employment at e^* , nontradable-sector employment at e^{N*} , tradable-sector employment at e^{T*} , housing price at p^{H*} , and number of commuters at zero. The gray dotted lines plot the equilibrium under a positive foreign demand shock of magnitude $\bar{C}_f^H$. The total employment in Regions 1 and 0 is $\bar{e}_1$ and $\bar{e}_0$, respectively, nontradable-sector employment is $\bar{e}_1^N$ and $\bar{e}_0^N$, tradable-sector employment is $\bar{e}_1^T$ and $\bar{e}_0^T$, housing price is $\bar{p}_1^H$ and $\bar{p}_0^H$, and the number of commuters is $\bar{\delta}$.

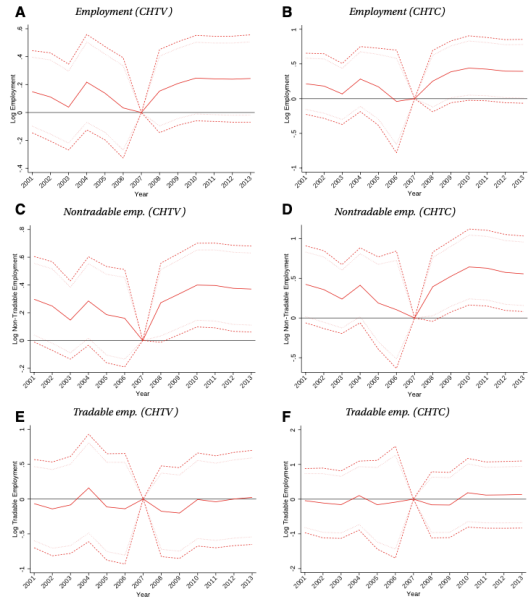


Figure 5
The dynamic effects of foreign Chinese housing demand on local employment
This figure plots the coefficients on the interaction terms between foreign Chinese housing transaction value or count (CHTV or CHTC) and a series of year dummies from regression Equations (8a) and (8b). The dependent variables are log total employment size (panels A–B), log nontradable-sector employment (panels C–D), and log tradable-sector employment (panels E–F). The left panels are based on CHTV as the main regressor, and the right panels are based on CHTC. The regressions control for pre-sample period ZIP code-level population, population density, education (population share with a bachelor's degree), and pretrends of income (1998–2001) and of the outcome variable (1996–2000), as well as county-year fixed effects, with 2007 used as the reference period. The 95% and 90% (in a lighter color) confidence intervals are drawn based on standard errors clustered at the ZIP code level.

5.7 Figure 5 and Tables 5–6: credibility and robustness

Read:

- Figure 5 shows flat pre-2008 event-study coefficients and positive post-2008 coefficients for total and nontradable employment.
- Table 5 shows no strong relationship between later Chinese inflows and earlier employment or income growth.
- Table 6 shows the employment and house-price results remain in the **2012–2013** subsample and after adding controls for **foreclosures**, **all-cash transaction share**, and **financial-sector exposure**.

Economic message. The paper works hard to show that the results are not just an artifact of the financial crisis or of Chinese buyers selecting already-rising neighborhoods.

Table 5
Reverse causality test

	Pre-policy shock period				Presample period			
	Employment		Income		Employment		Income	
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
$\ln(CHTV_{08-13})$	0.014 (0.012)		−0.005 (0.004)		−0.016 (0.012)		0.008 (0.009)	
$\ln(CHTC_{08-13})$		0.016 (0.013)		−0.006 (0.005)		−0.018 (0.013)		0.010 (0.010)
Controls	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
County FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
First-stage F -stat.	161	127	161	127	163	129	162	127
Obs.	628	628	628	628	629	629	627	627

This table reports regression results from Equations (9a) and (9b). The dependent variables are log change in total employment size and income from either the pre-China shock (2001–2007) or presample period (1996–2000 for employment and 1998–2001 for income). $CHTV_{08-13}$ ($CHTC_{08-13}$) denotes the log change in foreign Chinese housing transaction value (count) between 2008 and 2013, which is instrumented by the pre-sample period shares of ethnic Chinese population across ZIP codes. All regressions control for ZIP code-level population, population density, education (population share with a bachelor's degree), and county fixed effects. Standard errors, shown in parentheses, are clustered at the ZIP code level. * $p < .1$; ** $p < .05$; *** $p < .01$.

Table 6
Foreign Chinese housing demand, employment, and house prices: Controlling for financial crisis confounding factors

A. Total employment								
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
$\ln(CHTV) \times Post$	0.137** (0.066)		0.164** (0.065)		0.116* (0.062)		0.129** (0.059)	
$\ln(CHTV)$	0.052 (0.099)		0.001 (0.091)		0.058 (0.099)		0.030 (0.093)	
$\ln(CHTC) \times Post$		0.243** (0.102)		0.293*** (0.105)		0.233** (0.094)		0.251*** (0.094)
$\ln(CHTC)$		0.013 (0.117)		−0.083 (0.116)		0.000 (0.119)		−0.037 (0.112)
Standard Controls	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Additional Controls	–	–	Foreclosure	Foreclosure	All-cash Transactions	All-cash Transactions	Financial Sector	Financial Sector
County-year FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Postperiod	2012–13	2012–13	2008–13	2008–13	2008–13	2008–13	2008–13	2008–13
First-stage F -stat.	46	46	51	22	63	29	57	31
Obs.	2,410	2,410	3,877	3,877	4,166	4,166	4,166	4,166
B. Home prices (Zillow)								
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
$\ln(CHTV) \times Post$	0.120*** (0.022)		0.074*** (0.020)		0.102*** (0.020)		0.108*** (0.021)	
$\ln(CHTV)$	−0.019 (0.028)		−0.015 (0.025)		−0.020 (0.029)		−0.031 (0.027)	
$\ln(CHTC) \times Post$		0.190*** (0.041)		0.128*** (0.035)		0.184*** (0.039)		0.195*** (0.042)
$\ln(CHTC)$		−0.054 (0.037)		−0.050 (0.035)		−0.065 (0.041)		−0.085** (0.040)
Standard Controls	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Additional Controls	–	–	Foreclosure	Foreclosure	All-cash Transactions	All-cash Transactions	Financial Sector	Financial Sector
County-year FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Postperiod	2012–13	2012–13	2008–13	2008–13	2008–13	2008–13	2008–13	2008–13
First-stage F -stat.	46	25	58	21	64	24	59	25
Obs.	2,249	2,249	3,645	3,645	3,893	3,893	3,893	3,893

This table reports regression results from Equations (7a) and (7b) with log total employment size (panel A) and log Zillow Single-Family Home Value Index (panel B) as dependent variables. $CHTV$ ($CHTC$) denotes the foreign Chinese housing transaction value (count) instrumented by the weighted aggregate foreign Chinese housing transaction value (count) in California, with the pre-sample period shares of ethnic Chinese population across ZIP codes as weights. $Post$ is an indicator variable that takes the value 1 if the year is 2008 or after, and 0 otherwise. All regressions control for the pre-sample period ZIP code-level population, population density, education (population share with a bachelor's degree), an indicator variable for whether there is a college within a five-mile distance, and pretrends of income (1998–2001) and of the outcome variable (1996–2000). Columns 1–2 show the results for the post-global financial crisis period of 2012–2013. Columns 3–4 additionally control for the share of foreclosed homes in each ZIP code. Columns 5–6 additionally control for the share of all-cash house transactions in each ZIP code. Columns 7–8 additionally control for the size of the financial sector, measured as the share of financial sector employment in each ZIP code. All the additional controls are time-varying. Standard errors, shown in parentheses, are clustered at the ZIP code level. * $p < .1$; ** $p < .05$; *** $p < .01$.

5.8 Table 7: distributional consequences

Read:

- The number of **low-income** tax returns falls by **0.063%** per 1% increase in Chinese transaction value and by **0.112%** per 1% increase in Chinese transaction count.
- The number of **high-income** tax returns does not change significantly.

Table 7
Foreign Chinese housing demand and displacement of residents by income

	No. of Low-Income Tax Returns (under \$50,000)		No. of High-Income Tax Returns (above \$50,000)	
	(1)	(2)	(3)	(4)
$\ln(CHTV) \times Post$	-0.063*** (0.014)		-0.023 (0.031)	
$\ln(CHTV)$	0.027 (0.019)		0.127*** (0.044)	
$\ln(CHTC) \times Post$		-0.112*** (0.028)		-0.006 (0.051)
$\ln(CHTC)$		0.061*** (0.028)		0.133** (0.059)
Controls	Yes	Yes	Yes	Yes
County-year FE	Yes	Yes	Yes	Yes
First-stage F -stat.	53	22	53	22
Obs.	3,927	3,927	3,927	3,927

This table reports regression results from Equations (7a) and (7b) with the log number of income tax returns by households with income less than \$50,000 (columns 1–2) and greater than \$50,000 (columns 3–4) as dependent variables. $CHTV$ ($CHTC$) denotes the foreign Chinese housing transaction value (count) instrumented by the weighted aggregate foreign Chinese housing transaction value (count) in California, with the pre-sample period share of ethnic Chinese population across ZIP codes as weights. $Post$ is an indicator variable that takes the value 1 if the year is 2008 or after, and 0 otherwise. All regressions control for pre-sample period ZIP code-level population, population density, education (population share with a bachelor's degree), an indicator variable for whether there is a college within a five-mile distance, and pretrends of income (1998–2001) and of the outcome variable (1996–2000). Standard errors, shown in parentheses, are clustered at the ZIP code level. * $p < .1$; ** $p < .05$; *** $p < .01$.

Economic message. The paper's gentrification claim comes from here. The displacement is not uniform across residents; it is disproportionately borne by lower-income households.

5.9 External-validation calculation

Read:

- Using housing asset-to-equity ratios from the Survey of Consumer Finances and ACS, the paper backs out an implied elasticity of total employment with respect to **housing net worth** of about **0.43–0.61**.
- The corresponding elasticity for nontradable employment is about **0.55–0.78**, and for nontradable employment excluding construction it is about **0.35–0.50**.

Economic message. These magnitudes are broadly in line with Mian and Sufi (2014), which gives the paper an important external-validity check.

6 Short summary

- The paper measures foreign real estate capital inflows directly from U.S. housing transactions using a conservative three-step imputation algorithm.
- It documents a large post-2008 surge in Chinese housing purchases in California and a strong home-bias pattern toward neighborhoods with historical ethnic Chinese populations.
- Using a DID-IV design, it finds that Chinese housing purchases raise local employment, establishments, and house prices.
- The mechanism runs mainly through a **housing net worth channel** that boosts **nontradable-sector employment**.
- At the same time, the inflows **displace residents**, especially **low-income households**, and raise neighborhood average income.

- The paper's bottom line is that foreign housing capital inflows can simultaneously stimulate the local economy and intensify gentrification.

7 Conclusion

7.1 Core conclusion

The paper shows that foreign residential real estate capital inflows are economically important and can have large local real effects. In the California setting, the post-2008 surge in Chinese purchases raised employment and house prices in exposed neighborhoods.

7.2 Economic intuition

The main mechanism is balance-sheet based. When foreign buyers bid up local home prices, homeowners' housing wealth rises and this increases local spending on nontradable goods and services. But the same price increase also makes the neighborhood harder to afford, which pushes out residents.

7.3 Takeaway

This paper combines a novel measurement exercise. Its most important lesson is that foreign real estate inflows should be understood not only as an asset-market phenomenon, but also as a source of local demand stimulus with clear distributional costs.